

From Abstracts to Rooms: Poster Session Assignment by Semantic Similarity and Optimisation



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Declaration

I declare that the work presented in this dissertation is, to the best of my knowledge and belief, original and my own work. The material has not been submitted, either in whole or in part, for a degree at this, or any other university. Estimated word count is: **1860**

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Abstract

This is the beginning of the abstract. According to the Postgraduate Research Regulations 2022-23, it should not be any longer than 300 words. However, as your dissertation is to be submitted as part of a taught Master programme (which includes the MRes), there is no strict limit imposed on the abstract length. For general guidance on language, presentation, and ordering of your dissertation, refer to Appendix 2 (Pages 30-33) <https://bit.ly/2Q4H43I>.

Acknowledgements

Acknowledgements you may want to make. (this is not required when submitting your dissertation before your viva and you can add a dedication in your final dissertation after your viva if you wish.)

Contents

1	Introduction	1
1.1	Motivation	1
1.2	Problem Statement	1
1.3	Relevance	1
1.4	Proposed Approach	1
1.5	Contributions	1
1.6	Summary of Results	1
1.7	Did It Work?	1
1.8	Dissertation Structure	1
2	Literature Review	2
2.1	Conference Session and Poster Assignment	2
2.2	Representing Scientific Text as Vectors	3
2.3	Hierarchical Clustering of Documents	3
2.4	Capacitated Grouping and Clique Partitioning	4
2.5	Metaheuristics for Partitioning Problems	4
2.6	Benchmarking Algorithms	5
2.7	Summary and Positioning	5
3	Pipeline Overview	7
3.1	Design Goals	7
3.2	Architecture	7
3.3	Why This Decomposition	7
3.4	Notation and Terminology	7
3.5	Evaluation Strategy	7
4	Data Collection	8
4.1	Source and Method	8
4.2	The Dataset	8
4.3	Suitability and Limitations	8
4.4	The Paraphrase Evaluation Set	8

5	Embedding Model Selection	9
5.1	Representing Meaning as Vectors	9
5.2	Candidate Models	9
5.3	Experimental Design	9
5.4	Results	9
5.5	Discussion	9
6	Clustering	10
6.1	Hierarchical Agglomerative Clustering	10
6.2	Over-Clustering	10
6.3	Results	10
6.4	Discussion: Cohesion, Not Silhouette	10
7	Room Assignment	11
7.1	Problem Formulation	11
7.2	Exact Reference: Constraint Programming	11
7.3	Metaheuristics	11
7.4	Fairness Protocol	11
7.5	Statistical Comparison	11
7.6	Results	11
8	The Layout Tool	12
8.1	Purpose and Requirements	12
8.2	Architecture	12
8.3	Validation	12
8.4	Automatic Configuration	12
8.5	Outputs	12
8.6	Reproducibility and Portability	12
9	Results, Discussion and Analysis	13
9.1	The Final Layout	13
9.2	Per-Room Analysis	13
9.3	The Cost of Exact Fill	13
9.4	The Ranking of Methods Depends on the Budget	13
9.5	Objective and Similarity Do Not Always Agree	13
9.6	Interpreting the Magnitude	13
9.7	Against Expectations and Prior Work	13
10	Conclusion	14
10.1	Summary	14
10.2	Limitations	14

10.3 Future Work	14
Appendix A Supplementary Material	15
A.1 Room Capacities	15
A.2 Full Benchmark Tables	15
A.3 Cluster Examples	15
A.4 Code and Reproducibility	15
References	16

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Motivation

1.2 Problem Statement

1.3 Relevance

1.4 Proposed Approach

1.5 Contributions

1.6 Summary of Results

1.7 Did It Work?

1.8 Dissertation Structure

Chapter 2

Literature Review

2.1 Conference Session and Poster Assignment

The problem of organizing a scientific conference by computational means is not new. In an early study, Eglese and Rand (1987) surveyed the attendees of a 1985 conference to express which seminars they wished to attend, and then built a schedule that would allow the fulfillment of as many of these wants as possible. From that point on, the organizing of poster placements in a conference has become received a great deal of attention leading to the establishment of a large amount of operational research literature, where this problem has mainly been dealt as an integer linear problem. Research such as Stidsen, Pisinger, and Vigo (2018) describe a tool used to schedule the EURO-k conferences, events with more than two thousand presentations, by using optimization models to place related research buildings close to each other. Riquelme et al. (2022) focused on a "track-based" version of the problem—sorting choices into specific paths. Instead of finding an exact optimal solution, which is always computationally slow, they used a "simulated annealing heuristic," which is a smart shortcut algorithm. This shortcut found comparable solutions in a fraction of the time. Closest to our setting, Bulhões, Correia, and Subramanian (2022) define a general session scheduling problem whose objective is to maximize the similarity of papers put next to each other. They prove that this problem is NP-hard, and solve real conference instances with branch-and-cut and branch-cut-and-price methods. While this exact approach guarantees an optimal solution, it is computationally expensive, leaving room to develop faster methods that produce comparable results.

There is also a smaller line of research that finds topics by reading the actual paper texts, not by relying on pre-set labels or user preferences. Burke and Sabatta (2015) use Latent Dirichlet Allocation (LDA) directly to the submitted documents to learn the conference topics. They then schedule the parallel tracks so that simultaneous sessions cover the most different topics possible. By maximizing the distance between session topics, they ensure attendees are not forced to choose between two similar talks. In current practice, large

venues face the same task at much greater scale: For example, CVPR (CVPR, 2024) groups its accepted posters by similarity within each session. To decide which posters are similar, they use SPECTER embeddings and then reduce the size of those vectors down to a single line with one-dimensional t-SNE. Posters that land close together on that line are put in the same group. This is, to our knowledge, the closest real-world precedent for the problem we study; however, it is a projection heuristic rather than an explicit optimization, and the quality of the resulting layout is not measured.

Put together, the literature has largely pursued conflict avoidance and preference satisfaction. The topical coherence of a capacity-limited physical room, treated as an explicit optimization objective in itself, remains largely unexplored.

2.2 Representing Scientific Text as Vectors

Sentence embedding models map a text to a fixed-length vector such that semantically similar texts receive nearby vectors. Sentence-BERT introduced the now-standard approach of fine-tuning a pretrained transformer with siamese network structures, producing sentence vectors that can be compared directly with cosine similarity (Reimers and Gurevych, 2019). The model used later in this work, all-MiniLM-L6-v2, combines this training scheme with the MiniLM distillation method, which compresses a large teacher transformer into a six-layer student with 384-dimensional outputs (Wang et al., 2020). In contrast, a family of models exists specifically for scientific documents: SPECTER fine-tunes SciBERT on the citation graph, treating papers that cite one another as similar (Cohan et al., 2020); its successor SPECTER2 is trained on over six million citation triplets and attaches task-specific adapters (Singh et al., 2023); and Aspire models fine-grained similarity by matching individual sentences between papers (Mysore, Cohan, and Hope, 2022). The landscape is wider than the three models examined later: contrastive objectives such as SimCSE learn general-purpose sentence embeddings without any labelled pairs (**gao2021simcse**), and SciNCL replaces SPECTER’s discrete citation relations with sampling over citation-graph embeddings (**ostendorff2022neighborhood**). These models were trained on different signals and therefore encode different notions of what makes two papers similar, a distinction whose practical consequences we examine in Chapter 5.

2.3 Hierarchical Clustering of Documents

Clustering methods vary not solely in quality but in their structural properties, and these properties determine how well a method fits a given task. Hierarchical agglomerative clustering starts from every point as its own group and repeatedly merges the closest pair, producing a tree that can afterwards be cut at any level to yield any number of clusters; Ward’s method chooses each merge so that the increase of the within-group variance is

minimal, which favours compact groups (Ward, 1963). The k-means algorithm instead partitions the data into a number of groups that must be fixed in advance, and because it starts from a random initialization, its result can vary from run to run, which reduces reproducibility (MacQueen, 1967; Jain, 2010). Density-based methods such as HDBSCAN extract clusters of varying density from a hierarchy of density estimates (Campello, Moulavi, and Sander, 2013; McInnes, Healy, and Astels, 2017); a defining feature of this family is that points in low-density regions are labelled as noise and left outside every cluster. Finally, probabilistic topic models such as Latent Dirichlet Allocation (LDA) describe each document as a soft mixture of topics rather than assigning it to exactly one group (Blei, Ng, and Jordan, 2003).

In this work, clustering is judged by practical fit rather than benchmark accuracy: every poster must be placed (no unassigned outliers), the number of groups must be freely re-cut to match physical room capacities, and the outcome must be deterministic so that our layout heuristics remain stable. Chapter 6 derives the choice of hierarchical clustering directly from these three requirements.

2.4 Capacitated Grouping and Clique Partitioning

At its combinatorial core, our task is the clique partitioning problem: assign items to mutually exclusive groups so that the total similarity within groups is maximized. (Grötschel and Wakabayashi, 1989) introduced this formulation as a clustering model and proposed cutting plane methods for its solution. The problem was later shown to be NP-hard (Grötschel and Wakabayashi, 1990), and the geometric structure of its feasible solutions has been studied extensively (Oosten, Rutten, and Spieksma, 2001). Even with modern commercial solvers, an exact solution becomes impractical as instances grow, something that (Du et al., 2022) has also concluded through a direct comparison of models and solvers on clique partitioning benchmarks. Our problem in Chapter 7 adds two further restrictions to this core, namely room capacities and a minimum occupancy per room, and regardless of the added rules, it still doesn't remove its NP-hardness.

2.5 Metaheuristics for Partitioning Problems

For partitioning problems, a widely accepted set of metaheuristics exists. GRASP repeatedly builds a solution with a randomized greedy construction and then improves it with local search, keeping the best result over many restarts (Feo and Resende, 1995). Tabu search follows a single trajectory and uses a short-term memory of recently made moves to escape local optima (Glover and Laguna, 1997). Variable neighbourhood search alternates between descending to a local optimum and perturbing the solution with systematically growing shakes (Mladenović and Hansen, 1997). Memetic algorithms evolve a population of solutions

through recombination and mutation, with local search applied to the offspring (Moscatto and Cotta, 2003). These methods have been applied to clique partitioning itself, for example through iterated tabu search (Palubeckis, Ostreika, and Tomkevičius, 2014), and hybrid approaches additionally exist that embed an exact solver inside a heuristic loop, a design we will go back to in Chapter 7. The metaheuristic family is of course wider than these five; simulated annealing, for instance, has been applied in the conference scheduling setting itself as well (Riquelme et al., 2022).

2.6 Benchmarking Algorithms

All of the methods above are randomized, so a single run of each says little about which one is better. The established standard procedure is therefore to compare distributions of results over multiple independent runs, and to test the differences with non-parametric statistics, since the outcomes of heuristics cannot be assumed to follow a normal distribution. Demšar (2006) recommends exactly this design: the Wilcoxon signed-rank test for comparing two methods on paired results, and the Friedman test as an omnibus test when several methods are compared at once. Running many tests together creates a problem of its own: each test has a small chance of a false alarm, so across ten pairwise comparisons at least one “significant” result is quite likely to be fake. The procedure of Holm (1979) fixes this by making each individual test stricter, keeping the whole table at the usual five percent error level, and García and Herrera (2008) show how to apply it when every method is compared against every other, which is our case. Chapter 7 follows exactly this protocol; it is also why every method there receives the same time budget and the same random seeds, since the tests compare methods seed by seed, and that pairing is only fair if both methods faced identical conditions on each seed.

2.7 Summary and Positioning

Put together, the literature is divided into three paths. First, the operational research tradition schedules conferences well, but its objectives are preventing overlapping sessions and satisfying preferences, not the topical coherence of a room. Second, a smaller line of research reads the papers themselves to find topics, but stops at abstract groupings without rooms or seat capacities. Third, in practice, large venues already assign posters with embedding-based heuristics, without measuring the quality of the result. Every building block our pipeline needs exists on its own: sentence embeddings, hierarchical clustering, the clique partitioning problem, metaheuristics, and a statistical protocol for comparing them fairly. What no prior work does, to our knowledge, is combine them: a reproducible pipeline that takes a conference’s raw submissions and returns a topically coherent, physically feasible assignment of papers to capacity-limited rooms, with the solution methods given same amount of time

and computational power. That arrangement is what this dissertation provides.

Chapter 3

Pipeline Overview

3.1 Design Goals

3.2 Architecture

3.3 Why This Decomposition

3.4 Notation and Terminology

3.5 Evaluation Strategy

Chapter 4

Data Collection

4.1 Source and Method

4.2 The Dataset

4.3 Suitability and Limitations

4.4 The Paraphrase Evaluation Set

Chapter 5

Embedding Model Selection

5.1 Representing Meaning as Vectors

5.2 Candidate Models

5.3 Experimental Design

5.4 Results

5.5 Discussion

Chapter 6

Clustering

6.1 Hierarchical Agglomerative Clustering

6.2 Over-Clustering

6.3 Results

6.4 Discussion: Cohesion, Not Silhouette

Chapter 7

Room Assignment

7.1 Problem Formulation

7.2 Exact Reference: Constraint Programming

7.3 Metaheuristics

7.4 Fairness Protocol

7.5 Statistical Comparison

7.6 Results

Chapter 8

The Layout Tool

8.1 Purpose and Requirements

8.2 Architecture

8.3 Validation

8.4 Automatic Configuration

8.5 Outputs

8.6 Reproducibility and Portability

Chapter 9

Results, Discussion and Analysis

9.1 The Final Layout

9.2 Per-Room Analysis

9.3 The Cost of Exact Fill

9.4 The Ranking of Methods Depends on the Budget

9.5 Objective and Similarity Do Not Always Agree

9.6 Interpreting the Magnitude

9.7 Against Expectations and Prior Work

Chapter 10

Conclusion

10.1 Summary

10.2 Limitations

10.3 Future Work

Appendix A

Supplementary Material

A.1 Room Capacities

A.2 Full Benchmark Tables

A.3 Cluster Examples

A.4 Code and Reproducibility

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